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Langur Monkey Mother Loss: Profile Analysis with Multivariate Analysis of Variance for Separation Subjects and Controls

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Abstract. We present an analysis of observational data on the behavior profiles for two groups of infant langur monkeys *(Presbytis entellus):* one that had experienced mother loss, and the other, a control group, that had not undergone mother loss. Observations were analyzed in three 2-week time periods: 2 preseparation, 2 separation, and 2 postseparation periods for both groups. The profile analysis application of the multivariate analysis of variance (MANOVA) was employed to test the hypothesis that the two groups had similar patterns of change across time. For each group a vector of 5 difference scores was computed; then MANOVA was used to test the similarity of the mean vectors of difference scores for the two groups. Profiles for the separation and control infants were significantly different for some behaviors, and changes between some time periods were also significant.

Introduction

The Indian langur monkey *(Presbytis entellus),* an Old World Colobine, has been the subject of an investigation of response to mother-loss in a colony setting [Dolhinow, 1980]. Our primary goal here is the analysis of observational data on the separation and control groups of subject infants to determine whether the behavior profiles for each group were parallel over the three 2-week time periods of the separation experiment. The periods are preseparation, separation, and postseparation. Mothers of 6-to 8-month-

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old infants (mean age 32 weeks) were removed from the infants' social group for 2 weeks and then returned. Infants designated as controls were observed on the same schedule as the infants experiencing mother loss, but the control infants did not undergo maternal separation.

Responses to the separation experience have been presented elsewhere [Dolhinow, 1980]. Each infant was observed intensively during the 2 preseparation weeks to establish baseline levels for all behaviors. Separations were planned so that they did not overlap in time. When a mother was removed from the cage, her infant remained with the other members of its social group.

Other researchers have considered the frequency of three behaviors, play, locomotion, and vocalization, to be especially diagnostic of the effect of mother loss [Kaufman and Rosenblum, 1967a, b]. For the subject infant langurs, play scores for the 1st day of loss were not significantly affected because most of them did not perceive their loss until midday or later, and until that time their play scores were relatively normal. Vocalization and locomotion scores were markedly higher than normal for day 1 for separation subjects because once loss was perceived, each infant intently searched and called for its missing mother. When an infant began to concentrate on its search, play scores dropped to almost zero. Clearly, play was not a behavior of a distressed infant. By day 2 of separation, scores for play, vocalization, and locomotion were low. All infants sought care or at least contact with other adult females in the social group, and those that did obtain such substitute attention appeared to experience less distress than those that did not. Most separation subjects adopted adult females as substitutes for the missing mother by the 3rd day of separation. When mothers rejoined their infants' group, most infants that had been adopted remained with the adopted female. The few that had not obtained substitutes for the mother during separation returned to the mother when she was reintroduced at the end of the 2 weeks. In the long run, all infants maintained a strong relationship with some adult female, either the mother or a substitute for her. No infant became independent of an adult female by choice.

The postseparation period was marked by a gradual return to normative levels of behaviors such as play, vocalization, locomotion, grooming, and contact. Some behaviors recovered quickly while others took longer to reach normal levels. There was considerable variability among infants with respect to certain behaviors, as for example the amount of time they spent in contact with either the mother or her substitute. Patterns of response to loss were remarkably consistent among the separation subjects. They dem-

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onstrated a clear initial protest that lasted from 12 to 48 h. There followed varying degrees of depression or reduced activity, depending upon whether or not the infant was successful in obtaining a satisfactory substitute for its missing mother. 2 infants died during the early days of separation, and the deaths were clearly related to the devastation of loss. A 3rd infant died even though the mother was returned after an absence of only 2½ days. There is no question but that loss of the primary attachment object, the mother, is a potentially life-threatening event in the life of an immature langur monkey. The tremendous impact is primarily a psychological one, but this does not reduce its potential for damage because the emotional responses are quickly expressed physiologically [Reite et al., 1974, 1978a–c]. The great majority of infants, however, recovered from their temporary loss. In this report, we compare response to separation with the behavior of a group of control infants of the same age.

Materials and Methods

Animal Housing

The colony members live in two runways, each measuring 22.9 m in length, 6.0 m in width, and 4.6 m in height. Both have a concrete base, cyclone mesh fence on the front and top, corrugated metal back at one end, and a large heated sleeping room available to the animals at all times. All segments of the cages are equipped with running water, and heavy branches, tires, ropes, cargo nets, and other objects provide off-the-ground perches and play structures. The animals are fed in the morning after the runways are cleaned, and water is available at all times.

Subject and Controls

The subjects used in this research live in two social groups, and each infant has been observed since birth in its group of origin. 9 infants comprised the group of subjects that underwent mother loss [Dolhinow, 1980]. 7 animals served as controls: 2 females and 5 males. Every third birth was designated as a control. Observations for the preseparation period began when infants were approximately 32 weeks old. Table I lists the 7 controls, their sex, their mothers' and siblings' ages, and the infants' ages at the time of sampling, which were comparable to the ages of the separation subjects. All infants, controls and subjects, were observed prior to preseparation week 1 to determine whether they were eating sufficient solid food to survive if the mother were removed; all were adjudged potentially self-sufficient. Because every third birth was a control, the sex ratio of the control subjects was determined by the order of births in the colony. This was also the case with the sexes of the experimental subjects to which controls were compared. Controls lived in the same social groups as separation subjects, and it should be emphasized that each group was maintained unaltered except for births, deaths, and temporary removals of the separation subjects' mothers.

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Observation Procedures

Every animal in the two social groups was observed for six 10-min focal animal samples [Altmann, 1974] (a total of 1 h) per week to maintain baseline records. (Longer observations were made of the subject and control infants). Each sample focused on one animal, and all actions of, and interactions with that animal, were recorded. Both frequency and duration of behaviors were recorded. Observations were scheduled at different times of the day, and samples taken on any one day were not done continuously. Data for each sample included date, time of day, observer code, weather conditions, and proximity scores of the focal animal to all other animals in the social group at the beginning and end of each sample. Samples were divided into 1-min units for retrieval of behavior durations. The behavior repertoire used was designed for this and related research [Dolhinow, 1978].

Separations of mothers from subject infants were scheduled to avoid overlap; hence, every animal was present during the separation period except for the mother of a subject infant. Similarly, observations of control infants were scheduled so that no other separation or control sequence occurred at the same time. Each control was observed in three 2-week stages: preseparation, separation, and postseparation. The mean observation times per stage are as follows. During preseparation weeks 1 and 2, the infants were observed for

Table I. Control infants

Infant No.	Sex	Mother's age years	Siblings' ages years	Infant's age in weeks at 'separation'
1	f	8	2 3.50	32
2	f	15+	2 8	32
3	m	9	1.25	32
4	m	9	3 4.50 1.50	32
5	m	6	1.50	32
6	m	18+	2 8 11	33
7	m	11	2	33

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18 samples (3 h) weekly. During week 1 of the separation period, the controls were observed for 24 samples (4 h) weekly, and during week 2 of separation for 18 samples (3 h) weekly. Controls were also observed for 18 samples (3 h) weekly during postseparation weeks 1 and 2.

Data Analysis

Behavioral data were recorded on standardized recording sheets and transferred to computer disc by an interactive data entry program. The latter checked for inconsistencies and errors and allowed for corrections before the data were stored in a compact string form. A set of Fortran programs was then used to scan the data for individual behavior strings and produce summary tables. One program scanned for selected behavior sets and tallied the number of minutes in which the behaviors occurred at least once, whereas another sought out bouts of selected behaviors within a set of samples, and tallied the number of bouts occurring. A bout is a group of contiguous behavior sets such that the first behavior set involves the focal animal (and possibly other animals). The bout is terminated by a behavior set that does not contain the selected behavior with the same animals. These programs gave detailed printouts for each animal, and described the mode of the behavior (whether the animal was actor, solitary, recipient, or participant). A third program tallied the number of minutes an animal was observed in any of a set of selected behaviors, and printed out the total durations for each of the animals involved.

For each animal, 10-min samples were organized into blocks for each week of the 6-week separation schedule. Within a time period, the actual frequency of bouts was divided by the total number of samples taken during the time period, giving a bout per sample score. Likewise, in behaviors measured as present or absent during a sample minute, the number of minutes in which the behavior occurred was divided by the total number of minutes in the time period, thus giving an estimate of the percentage of sample time in which the behavior occurred. Then, in each behavior these values were assembled into 16×6 matrices representing the scores of the 16 infants (9 separation, 7 control) across the 6 weeks of the separation schedule. The matrices were put into the SPSS multivariate analysis of variance (MANOVA) subprogram, which performed the statistical analysis. Data entry and processing were done on the University of California Quantitative Anthropology Laboratory's Data General MV/8000 and on the University of California Computer Center's CDC 6400 at Berkeley.

Data are available on the occurrence, frequency, or duration of every behavior in the repertoire of the langur monkey, but only those considered most diagnostic of the impact of mother loss are presented here. These behaviors include approaches to all animals by the infant, locomotion, vocalization, social play, proximity to other animals, grooming infant as receiver, mouthing and/or manipulating, approaches to the infant, and frequency of eating. Other behaviors such as grooming with infant as actor, protective behaviors, rejections, and threats to infants, were not tested with the multivariate techniques used herein because the data matrices contained too many zero scores.

Our primary goal in analyzing these data was to determine whether profiles for the separation and control subject groups were parallel over the 6 weeks covered by preseparation, separation, and postseparation. The profile analysis application of MANOVA [Morrison, 1976, p. 205] was employed to test the hypothesis that the two groups had similar patterns of change across time. This is equivalent to testing for a group $\times$ week interaction with conventional analyses of variance. For each subject a vector of 5 different scores was

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computed: preseparation week 1 minus preseparation week 2, preseparation week 2 minus separation week 1, and so forth. Then MANOVA was used to test similarity of mean vectors of different scores for the two groups. Separate analyses were carried out for each behavior.

The advantage of using MANOVA in this analysis over conventional mixed-model analyses of variance is that fewer assumptions are required about the data. The mixed-model approach required that population covariance matrices for each level of nonrepeated factor be equal, and that differences between means for all pairs of levels on the repeated factor be equal [Winer, 1971]. The MANOVA approach does not require that these conditions be met, and is therefore preferable for many applications.

Two statistics are generated by MANOVA: Wilk's Lambda and Rao's F. Wilk's Lambda statistic varies between zero and one; the smaller its value the greater the difference in the profiles on the separation and control groups. Rao's F statistic is based on the magnitude of Wilk's Lambda, and can be referred to the F distribution to obtain a probability level for the similarity of profiles test.

Table II. Results of profile analysis with multivariate analysis of variance for eleven behaviors

Behavior	Wilk's Lambda	Rao's F[1]	Probability
Approaches by infant	0.219	7.12	0.004[2]
Locomotion	0.282	5.08	0.014[2]
Vocalization	0.316	4.34	0.023[2]
Social play	0.198	8.13	0.003[2]
Play solitary	0.636	1.15	0.398
Proximity time	0.380	3.27	0.052
Contact time	0.543	0.854	0.543
Groom time, infant receiver	0.680	0.941	0.495
Mouth, manipulate	0.742	0.696	0.639
Approaches to infant	0.698	0.866	0.536
Eat	0.698	0.866	0.536

[1] Degrees of freedom for Rao's F = 5, 10.
[2] Statistically significant.

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Results

Table II presents the results of profile analyses with multivariate analysis of variance for eleven behaviors under consideration. Profiles for separation and control infants were significantly different for the following behaviors: approaches by the infant to all other animals in the group, locomotion, vocalization, and social play. Examination of Wilk's Lambda statistics reveals that differences between the profiles for the two groups were greatest for social play, followed by approaches by the infant, and then by locomotion and vocalization.

MANOVA generates a canonical variate, which represents a linear combination of difference scores that maximally discriminates between the two groups. The magnitudes of correlations between different scores and canonical variate indicate between which points in time the profiles for experimental and control groups are most dissimilar. These correlations are presented in table III.

The profiles for the two groups in approaches by the infant, for example, are most dissimilar between preseparation week 2 minus separation week 1. Locomotion, vocalization, and social play, all show the most dissimilarities in the preseparation week 2 minus separation week 1, and separation week 1 minus separation week 2 scores.

Figures 1–4 present the differences between means for consecutive time periods for each of the four behaviors in table III. Bars in these figures

Table III. Magnitudes of correlations between different scores and canonical variate

Time intervals	Different scores	Approaches by infant	Locomotion	Vocalization	Social play
1	PRW 1 minus PRW2	0.27	0.05	0.21	0.03
2	PRW 2 minus SPW 1	0.48	0.76	0.73	0.62
3	SPW 1 minus SPW 2	0.12	0.55	0.84	0.81
4	SPW 2 minus PSW 1	0.05	0.15	0.45	0.17
5	PSW 1 minus PSW 2	0.09	0.11	0.48	0.30

PRW = Preseparation week; SPW = separation week; PSW = postseparation week.

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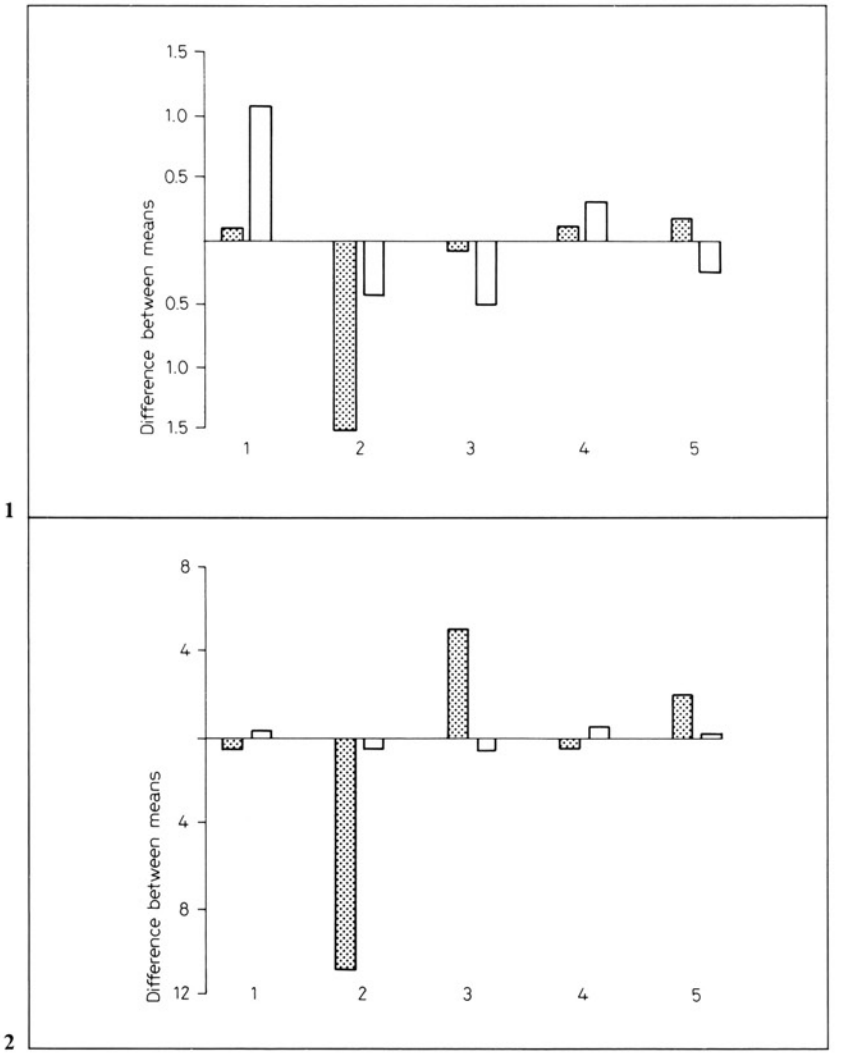


Fig. 1. Differences between means for consecutive time periods for approaches by the infant to all animals in the group. The shaded bars are experimental subjects, and the open bars are control subjects. The 5 pairs of bars represent the 5 time intervals given in table III. Bar 1 is the different score preseparation week 1 minus preseparation week 2; bar 2, preseparation week 2 minus separation week 1; bar 3, separation week 1 minus separation week 2; bar 4, separation week 2 minus postseparation week 1; and bar 5 post-separation week 1 minus postseparation week 2.

Fig. 2. Differences between means for consecutive time periods for locomotion by the infants. Explanation of symbols and bars as in figure 1.

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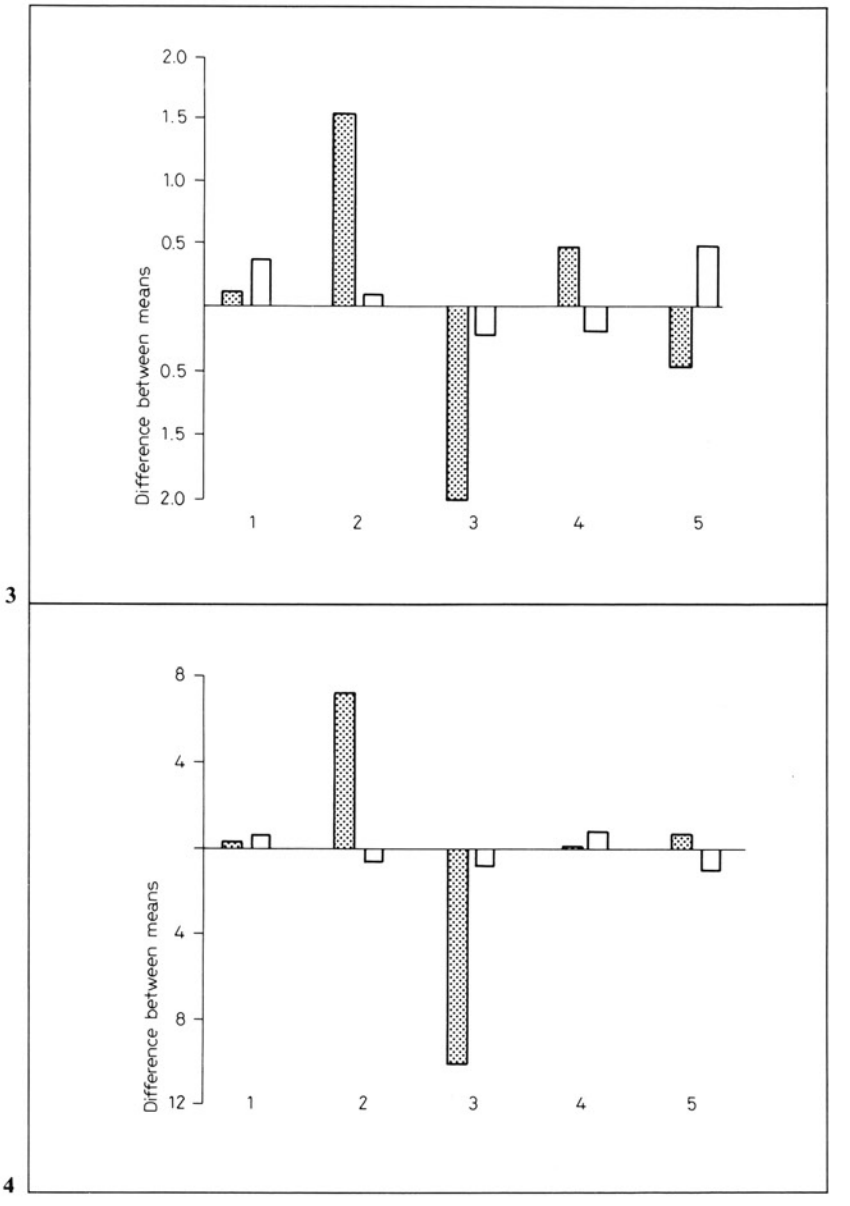


Fig. 3. Differences between means for consecutive time periods for vocalization by the infants. Explanation of symbols and bars as in figure 1.

Fig. 4. Differences between means for consecutive time periods for social play by the infants. Explanation of symbols and bars as in figure 1.

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represent differences between mean scores for consecutive weeks. These values were obtained by subtracting the mean score for a week from the mean score of the preceding week. For example, as illustrated in figure 1, the mean score for controls increased between preseparation week 1 and preseparation week 2; this is represented by a bar above the horizontal with a magnitude equal to the difference between means. Between preseparation week 2 and separation week 1, the mean approach score for the controls decreased; this is represented by a bar below the horizontal.

In general, there is much change in the behaviors of the experimental group subject to the treatment, and not in the behaviors of the controls. This may be seen by inspection of figures 1–4. As is to be expected in a sample of this size, there are random fluctuations in the behavior of every group of animals – control as well as experimental. These fluctuations are reflected in mean scores for specific periods of time. However, the highly significant differences in behavior between control and experimental groups are readily apparent when the figures are inspected. The greatest differences between the approach (fig. 1), locomotion (fig. 2), vocalization (fig. 3), and social play behaviors (fig. 4) of the control and the separation subjects occur during the 2 weeks of the separation period.

Discussion

Infant reaction to mother loss has been studied using several species of macaques, and the results, although variable [Lewis et al., 1976; Mineka and Suomi, 1978], generally picture a depressive response of varying intensity [Abrams, 1969; Hansen, 1966; Harlow and Suomi, 1974; Harlow et al., 1970; Hinde, 1974; Kaufman, 1977; Kaufman and Rosenblum, 1967a, b, 1969; Kaufman and Stynes, 1978; Mitchell et al., 1967; Preston et al., 1970; Reite et al., 1974; Rosenblum and Plimpton, 1981; Seay et al., 1962; Seay and Harlow, 1965; Spencer-Booth and Hinde, 1967, 1971]. There are important differences among these studies of attachment and loss, as for example, in the length of separation, size and composition of social groups, length of time the social groups were in existence before separation, and ages of infants subject to loss. Resulting variability in responses is not, therefore, unexpected or unexplainable.

The study reported upon here is the only study of mother loss and attachment bonds in an Old World Colobine monkey. It was undertaken to

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provide comparative information on the attachment behavior of a monkey for which prolonged and intimate contact with many adults in the social group is the normal situation from birth. In general, the data we have obtained as a result of experimental separation of mothers from their infants have been comparable to the results of Kaufman and his associates on two species of macaques.

The experimental and control infants in the present report were reared from birth by their mothers in two stable social groups composed of related and unrelated individuals of both sexes and different ages. The two groups were constituted and have been maintained to be comparable in size and composition to those observed in the wild in India and Nepal [Bishop, 1975; Boggess, 1976; Curtin, 1975; Hrdy, 1977; Jay, 1963, 1965; Mohnot, 1971; Sugiyama, 1964, 1965; Vogel, 1970; and see also review in Roonwall and Mohnot, 1976].

The results of our mother loss experimentation are clear and strikingly consistent for all subjects. The impact of loss is strong, even for infants that succeed in obtaining a substitute for their missing mother. All infants that survived the first 5 days of their loss eventually recovered; that is, all behaviors designated as sensitive to the loss reaction returned to normative range for individuals of that age.

For 9 separation subjects reported upon elsewhere [Dolhinow, 1980], a one-way analysis of variance for repeated measures revealed significant differences between days for vocalizations, locomotion, play, and approach to all animals. Pairwise comparisons revealed significant differences between preseparation week 2 and separation day 2 for vocalization, locomotion, play, and approach to all animals; and between day 1 and 2 of separation as well. However, between preseparation week 1 and postseparation week 2 the pairwise comparisons showed no significant differences for vocalization or locomotion, although they did show significant differences for both play and approach to all animals.

In the present study, profile analysis application of multivariate analysis of variance to the observational data from experimental and control subjects revealed significant differences between the two groups of subjects over the 6 time periods. Play, vocalization, and locomotion, as anticipated, were very sensitive to the distress perceived by the infant upon loss of its mother [Dolhinow, 1980]. Table II illustrates that these three behaviors and approaches to all animals by the infants were significantly different in profile for the separation and control subject groups. The profiles were most different for play, and less so for approaches by the infant, for locomotion,

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and vocalization. Social play decreased to almost zero for the separation subjects. Play bouts recorded were almost entirely initiated by other infants. The great majority of invitations were either ignored or actively discouraged as bereft infants attempted to reestablish contact with the mother, or to gain and maintain contact with a substitute for here. There was no significant difference in solitary play profiles between controls and experimental infants, since solitary play is very infrequent at this age. Infant approaches to other members of its group increased for separation subjects as they searched for the missing mother. Similarly, as they tried to regain contact with the loss object, separation infants wandered and their locomotion scores increased. As they searched, they usually called, and hence vocalization scores increased dramatically on the day of loss and remained higher than normal for some days.

Because langur monkeys characteristically rest in proximity or in contact with others, there were no significant differences in the profiles of proximity time or contact scores for the separation and control infants. The separation subjects neither isolated themselves from others nor maintained higher contact scores than infants with the mother present. Langurs are exceedingly tolerant of passive contact, and group members did not appear to discriminate against separation subjects. The only exception occurred when nonseparated infants defended their place at their mother's ventrum against intrusion by a separation subject in search of a substitute for its own mother. The profiles for grooming of, and approaches to, the infant were not different for controls and separation subjects. Adults did not respond to a separation infant's visible and audible signs of distress by either approaching or grooming it more often than they did an infant whose mother had not been removed from the group. Infant relief, if it was to come at all, came as a result of infant initiative, and not from an acting out of adult concern. In short, unless a separation subject succeeded in stimulating attention from a specific adult, female adults treated infants suffering loss as they did control infants. Infants of separation age do not mouth or manipulate objects as do younger infants, nor was this a significant aspect of behavior for separation subjects. Eating did not increase or decrease markedly for the infants of either group during the time periods investigated. Although separated infants ate as often, we do not know the quantity of food they actually ingested, and think that they probably consumed less when they were distressed. One infant (not included in the sample of separation infants because her separation was not completed) died, and the apparent cause of death upon autopsy was starvation [paper in prep.].

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It is important to note that behaviors such as protection, rejection, threat, and infant grooming of others, occurred so infrequently in both groups that they could not be analyzed with multivariate statistics. Although separation subjects sought contact and attention, sometimes rather aggressively and loudly, this behavior did not appreciably increase aggression directed toward them. Other langurs may not have been solicitous of distressed infants, but at least they were tolerant of them.

Table III illustrates that in the 6 weeks the profiles for the two groups, separation and controls, were most different. Changes between preseparation week 2 and separation week 1 in approaches to all animals by the infant in locomotion, vocalization and play, were markedly different for the two groups. The two also differed between separation week 1 and separation week 2 in locomotion, vocalization, and social play. Between other time periods, the differences were less dramatic.

Although langur monkey infants have much contact with other adults in the group besides the mother, there is no ambiguity as to which adult is paramount to the infant's life strategy. It is the mother, and when she is missing the infant spends great effort to regain her or, failing that, to replace her. It is apparent from the focus of an infant's distress on the loss of a specific individual, its mother, that it is strongly attached to her. This focus is one measure often used in determining whether a prior attachment has existed [Ainsworth, 1972].

Summary

An analysis is presented of observational data on the behavior profiles for two groups of infant langur monkeys. One group experienced loss of the mother for 2 weeks; the other, a control group of infants of the same age, did not experience mother-loss. Observations were analyzed in three 2-week time periods: 2 preseparation, 2 separation, and 2 post-separation for both groups of infants. The profile analysis application of MANOVA was employed to test the hypothesis that the two groups had similar patterns of change across time. For each group a vector of 5 different scores was computed. MANOVA was then employed to test the similarity of the mean vectors of different scores for the two groups of infants. Profiles for the separation and control infants were significantly different for approaches by the infant to all other animals in the group, and for locomotion, vocalization, and social play. Changes between preseparation week 2 and separation week 1 in the infant's approaches to all animals, and in play, vocalization, and locomotion, were greatly different for the two groups. They also differed between separation week 1 and separation week 2 in locomotion, vocalization, and social play. Differences were less dramatic between other time periods.

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